



Time Series Analysis

Seminar 0: Introduction: components and exponential smoothing



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Acronyms used in this material

- ▣ **AI: Artificial Intelligence**
- ▣ **CV: Cross-Validation**
- ▣ **LLM: Large Language Model**
- ▣ **MA: Moving Average**
- ▣ **MA-5: 5-period Moving Average**
- ▣ **MAE: Mean Absolute Error**
- ▣ **MAPE: Mean Absolute Percentage Error**
- ▣ **MSE: Mean Squared Error**
- ▣ **RMSE: Root Mean Squared Error**
- ▣ **SES: Simple Exponential Smoothing**

Seminar Outline

- ▣ **Multiple Choice Quiz** – Knowledge check
- ▣ **True/False** – Conceptual checks
- ▣ **Calculation Exercises** – Applied practice
- ▣ **AI-Assisted Exercise** – Critical thinking
- ▣ **Summary** – Key takeaways

Quiz 1: Time Series Basics

Question

Which of the following is NOT a characteristic of time series data?

Answer choices

- (A) Observations are ordered in time
- (B) Consecutive observations are usually correlated
- (C) Observations are independent and identically distributed
- (D) Data has a natural temporal ordering

Answer on next slide...

Quiz 1: Answer

Answer: C – Observations are independent and identically distributed

❑ **Question:** Which is NOT a characteristic of time series data?

Answer choices

- (A) Observations are ordered in time ✗
- (B) Consecutive observations are usually correlated ✗
- (C) **Observations are independent and identically distributed** ✓
- (D) Data has a natural temporal ordering ✗

- ❑ Time series observations are **dependent** (autocorrelated), not independent
- ❑ The i.i.d. assumption is fundamental to cross-sectional analysis but is **violated** in time series
- ❑ This temporal dependence requires **specialized methods**



Quiz 2: Decomposition

Question

When should you use multiplicative decomposition instead of additive?

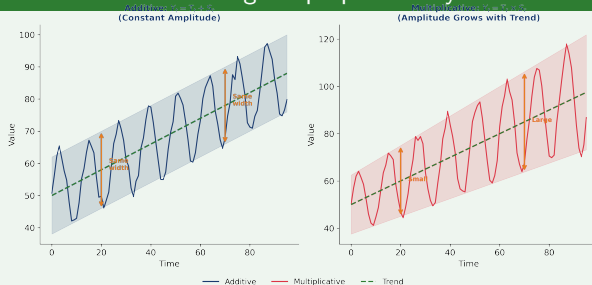
Answer choices

- (A) When the seasonal pattern has constant amplitude
- (B) When the variance of the series is stable over time
- (C) When seasonal fluctuations grow proportionally with the level
- (D) When the series has no trend component

Answer on next slide...

Quiz 2: Answer

Answer: C – When seasonal fluctuations grow proportionally with the level



- ☐ **Multiplicative:** $X_t = T_t \times S_t \times \varepsilon_t$ — seasonal amplitude scales with the level
- ☐ **Additive:** $X_t = T_t + S_t + \varepsilon_t$ — constant amplitude

Quiz 3: Exponential Smoothing

Question

In Simple Exponential Smoothing with $\alpha = 0.9$, what happens?

Answer choices

- (A) Forecasts are very smooth and stable
- (B) Recent observations have very little weight
- (C) Forecasts react quickly to recent changes
- (D) The forecast is essentially a long-term average

Answer on next slide...

Quiz 3: Answer

Answer: C – Forecasts react quickly to recent changes

- ▣ With $\alpha = 0.9$: $\hat{X}_{t+1} = 0.9X_t + 0.1\hat{X}_t$
- ▣ **High** α (e.g. 0.9): 90% weight on the last observation
 - ▶ Forecasts very responsive to new data
- ▣ **Low** α (e.g. 0.1): smoother, more stable forecasts
 - ▶ Averages over more history

Quiz 4: Moving Averages

Question

A centered moving average of order 5 (MA-5) uses which observations to estimate the trend at time t ?

Answer choices

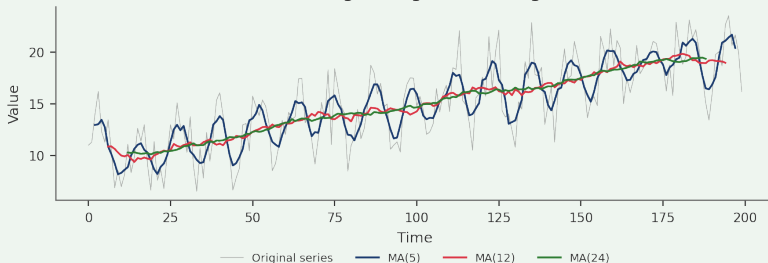
- (A) $X_t, X_{t+1}, X_{t+2}, X_{t+3}, X_{t+4}$
- (B) $X_{t-4}, X_{t-3}, X_{t-2}, X_{t-1}, X_t$
- (C) $X_{t-2}, X_{t-1}, X_t, X_{t+1}, X_{t+2}$
- (D) X_{t-1}, X_t, X_{t+1}

Answer on next slide...

Quiz 4: Answer

Answer: C – $X_{t-2}, X_{t-1}, X_t, X_{t+1}, X_{t+2}$

Moving average: smoothing



- ☐ **Centered MA:** uses $(k - 1)/2$ observations on each side of t
- ☐ **MA-5:** 2 before + t + 2 after $\Rightarrow$ larger window = smoother

Quiz 5: Forecast Evaluation

Question

Which metric is most appropriate for comparing forecast accuracy across series with different scales?

Answer choices

- (A) Mean Absolute Error (MAE)
- (B) Root Mean Squared Error (RMSE)
- (C) Mean Absolute Percentage Error (MAPE)
- (D) Mean Squared Error (MSE)

Answer on next slide...

Quiz 5: Answer

Answer: C – Mean Absolute Percentage Error (MAPE)

- ▣ $MAPE = \frac{100}{n} \sum \left| \frac{e_t}{X_t} \right|$ expresses errors as **percentages**.
- ▣ MAE, RMSE, MSE are **scale-dependent** (units of X_t)
- ▣ MAPE is **scale-independent** (always in %)
- ▣ Caveat: MAPE becomes unstable when $X_t \approx 0$

 TSA_ch0_forecast_eval

Quiz 6: Cross-Validation

Question

Why can't we use standard k-fold cross-validation for time series?

Answer choices

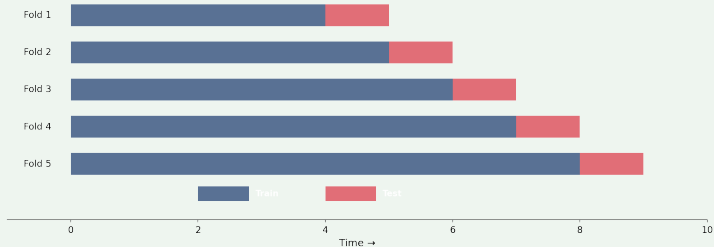
- (A) Time series data is too small
- (B) It would violate temporal ordering (future predicting past)
- (C) Cross-validation is always invalid
- (D) Time series doesn't need validation

Answer on next slide...

Quiz 6: Answer

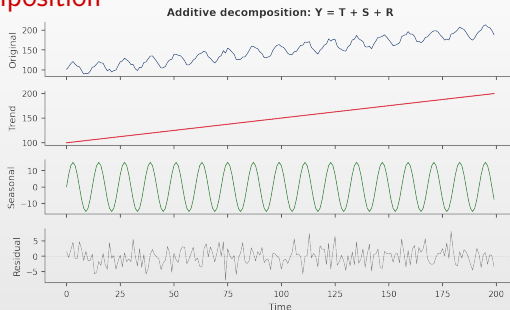
Answer: B – It would violate temporal ordering

Time Series Cross-Validation (Walk-Forward)



- ☐ **Principle:** future data cannot be used to predict the past! Rolling/expanding window CV is recommended.

Visual: Time Series Decomposition



Decomposition Components

☐ **Trend:** long-term movement **Seasonality:** periodic pattern **Residual:** random noise

True or False? — Questions

Statement	T/F?
1. SES forecasts are flat (constant for all horizons).	?
2. RMSE penalizes large errors more than MAE.	?
3. Multiplicative decomposition requires positive data.	?
4. A larger α means more smoothing.	?
5. The test set is used for hyperparameter tuning.	?
6. Seasonal naive uses the value from one season ago.	?
7. MAPE can be infinite if actual values are zero.	?

True or False? — Answers

Statement	T/F	Explanation
1. SES forecasts are flat (constant for all horizons).	T	No trend
2. RMSE penalizes large errors more than MAE.	T	Squared errors
3. Multiplicative decomposition requires positive data.	T	Cannot $\times$ negative
4. A larger α means more smoothing.	F	Large α = less smooth
5. The test set is used for hyperparameter tuning.	F	Use validation!
6. Seasonal naive uses the value from one season ago.	T	$\hat{X}_{t+h} = X_{t+h-m}$
7. MAPE can be infinite if actual values are zero.	T	Division by zero

Exercise 1: Simple Exponential Smoothing

Problem

- ▣ **Data:** $X = [10, 12, 11, 14, 13]$ with $\alpha = 0.3$, $\hat{X}_1 = 10$
- ▣ **Calculate:** a) Forecasts $\hat{X}_2$ through $\hat{X}_6$; b) MAE and RMSE
- ▣ **Formula:** $\hat{X}_{t+1} = \alpha X_t + (1 - \alpha)\hat{X}_t$

Solution

t	1	2	3	4	5	6
X_t	10	12	11	14	13	?
$\hat{X}_t$	10	10	10.6	10.72	11.70	12.09

- ▣ **MAE** = 1.745 **RMSE** = 2.04

Exercise 2: Error Metrics

Problem

- ▣ **Data:** $X = [100, 110, 105, 120]$, $\hat{X} = [95, 108, 110, 115]$
- ▣ **Calculate:** MAE, MSE, RMSE, MAPE

Solution

- ▣ **Errors:** $e = [5, 2, -5, 5]$
- ▣ **MAE** $= (|5| + |2| + |-5| + |5|)/4 = 4.25$
- ▣ **MSE** $= (25 + 4 + 25 + 25)/4 = 19.75$
- ▣ **RMSE** $= \sqrt{19.75} = 4.44$
- ▣ **MAPE** $= 25 \times (0.05 + 0.018 + 0.048 + 0.042) = 3.95\%$

Exercise 3: Seasonal Indices

Problem

- ▣ **Data:** Seasonal indices: $S = [0.85, 1.05, 0.90, 1.20]$, Trend Q4: $T = 1000$
- ▣ **Calculate:** a) Verify normalization. b) Q4 forecast. c) Deseasonalize $X_{Q4} = 1150$

Solution

- ▣ **a) Normalization:** $\sum S_i = 0.85 + 1.05 + 0.90 + 1.20 = 4.00 \checkmark$
- ▣ **b) Forecast:** $\hat{X}_{Q4} = 1000 \times 1.20 = \mathbf{1200}$
- ▣ **c) Deseasonalization:** $X_{deseasonalized} = 1150/1.20 = \mathbf{958.33}$ (below trend)

AI Exercise: Critical Thinking

Prompt to test in ChatGPT / Claude / Copilot

"Using yfinance, download adjusted close prices for SPY from 2015 to 2024. Apply seasonal decomposition (additive and multiplicative) and compare the results visually. Then split the data into training set (2015–2023) and test set (2024). Fit three exponential smoothing models (SES, Holt, Holt-Winters) on the training set and evaluate them on the test set using RMSE and MAPE. Which model is best? Show the plots and comparison table."

Exercise:

1. Run the prompt in an LLM of your choice and critically analyze the response.
2. Does the AI choose additive or multiplicative decomposition? Is the choice justified?
3. How does it evaluate models — does it use train or test RMSE?
4. Check the smoothing parameters (α , β , γ). Are values near 1.0 problematic?
5. Does the code properly split data into train/test, or does it evaluate on training data?

Warning: AI-generated code may run without errors and look professional. *That does not mean it is correct.*

Summary: Chapter 0

Key Concepts

1. **Time series:** temporally ordered observations, with dependence (autocorrelation)
2. **Decomposition:** additive ($X_t = T_t + S_t + \varepsilon_t$) vs multiplicative ($X_t = T_t \times S_t \times \varepsilon_t$)
3. **Exponential smoothing:** SES, Holt, Holt-Winters — parameter α controls reactivity
4. **Forecast evaluation:** MAE, RMSE, MAPE — the choice depends on context
5. **Seasonality:** seasonal indices, forecasting and deseasonalization

Questions?

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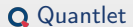
Online Resources and Code

- ▣ **Quantlet**: <https://quantlet.com> — Code repository for statistics
- ▣ **Quantinar**: <https://quantinar.com> — Quantitative methods learning platform
- ▣ **GitHub TSA**: https://github.com/QuantLet/TSA/tree/main/TSA_ch0 — Python code for this seminar

Thank You!

Questions?

Seminar materials are available at: <https://danpele.github.io/Time-Series-Analysis/>



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